

Metropolitan University, Sylhet
Department of Computer Science and Engineering
Spring Semester Final Examination – 2024

Program: B.Sc. in CSE; Batch – 58

Course: MAT 135: Matrices, Complex Variable and Fourier Analysis

Time: 2 Hours

Marks: 40

Answer any four questions. All parts of each questions must be answered sequentially. Figures in the right margin indicate full marks.

Group – A

(Answer any two questions)

- ✓ (a) Determine whether or not the vectors 5
- $$u_1 = (1,1,1), \quad v = (1,2,3), \quad w = (2,-1,1)$$
- are linearly dependent.
- (b) Solve the following system of linear equation by using matrices 5
- $$\begin{aligned} x + y + z &= 6 \\ x - y + z &= 2 \\ 2x + y - z &= 1 \end{aligned}$$
2. (a) Define Characteristic Polynomial, Eigenvalues and Eigenvectors of a Matrix. Show that, 5
- $$A = \begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix}$$
- is a root of its characteristic polynomial.
- (b) Find all eigenvalues and corresponding eigenvectors of matrix 5
- $$A = \begin{pmatrix} 4 & 2 \\ 3 & -1 \end{pmatrix}$$
- ✓ (a) Suppose $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$ then prove that, 5
- $$\begin{aligned} \text{(i)} \quad z_1 z_2 &= r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)] \\ \text{(ii)} \quad \frac{z_1}{z_2} &= \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)] \end{aligned}$$
- (c) State and prove the De Moivre's theorem. 5

Group – B

(Answer any two questions)

- ✓ (a) Suppose z_1 and z_2 are two complex numbers, show that, $|z_1 + z_2| \leq |z_1| + |z_2|$. 5
- (b) Define Harmonic Function and show that $u = e^{-x}(x \sin y - y \cos y)$ is harmonic. 5
- ✓ (a) Define Fourier series of a function $f(x)$ with the period of length 2π . 2
- (b) Find the Fourier series expansion of the following below with period: 8

$$f(x) = \begin{cases} -x; & -2 \leq x \leq 0 \\ x; & 0 \leq x \leq 2 \end{cases}$$

- 6 Use Finite Fourier Sine Transform to solve the following differential equation: 10

$$\begin{aligned} \frac{\partial U}{\partial t} &= \frac{\partial^2 U}{\partial x^2}; \text{ where,} \\ U(0, t) &= U(\pi, t) = 0; \\ U(x, 0) &= 2x; 0 < x < \pi; t > 0. \end{aligned}$$

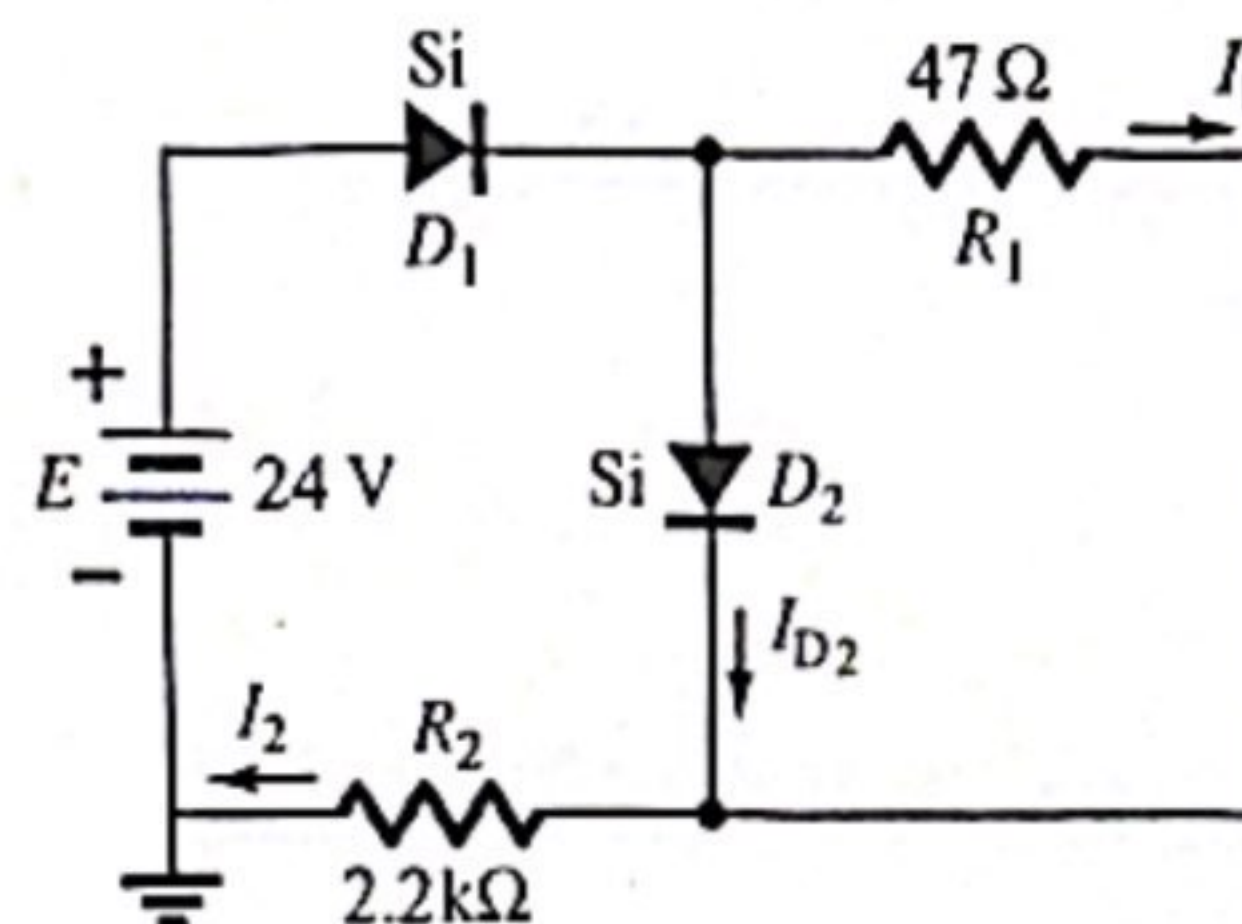
Metropolitan University, Sylhet
 Department of Computer Science and Engineering
 Spring Semester Final Examination – 2024
 Program: B.Sc. in CSE; Batch: 58
 Course: CSE-131: Basic Electronics Engineering

Time: 2 Hours

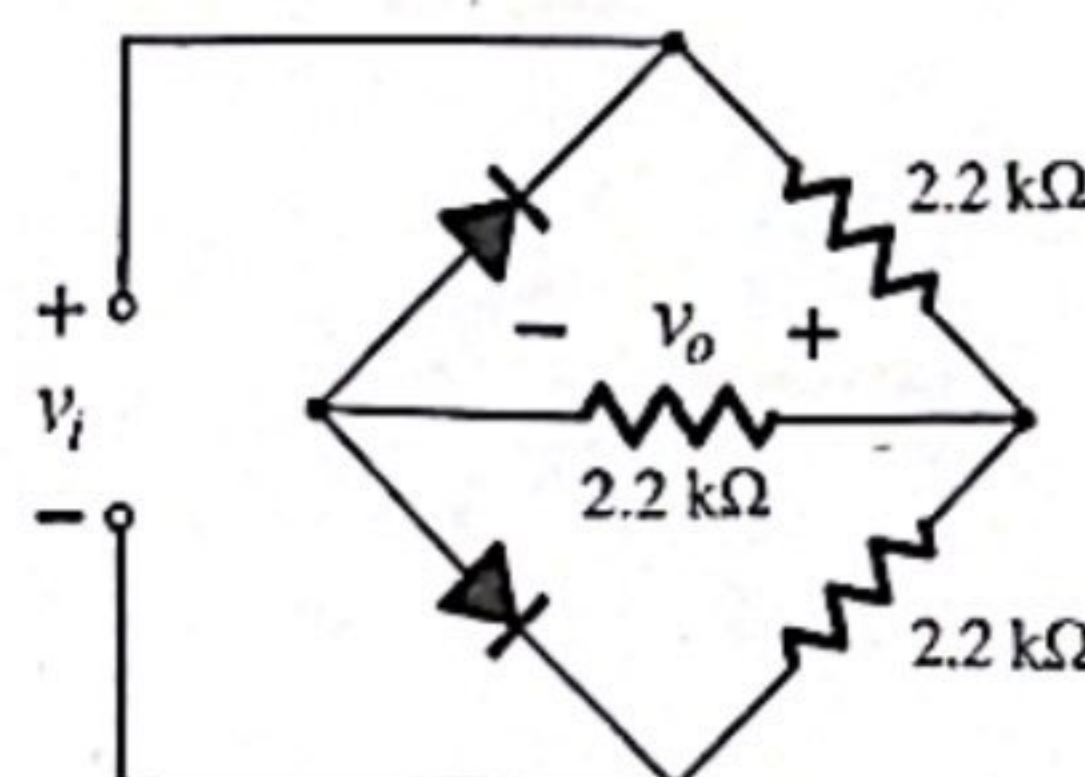
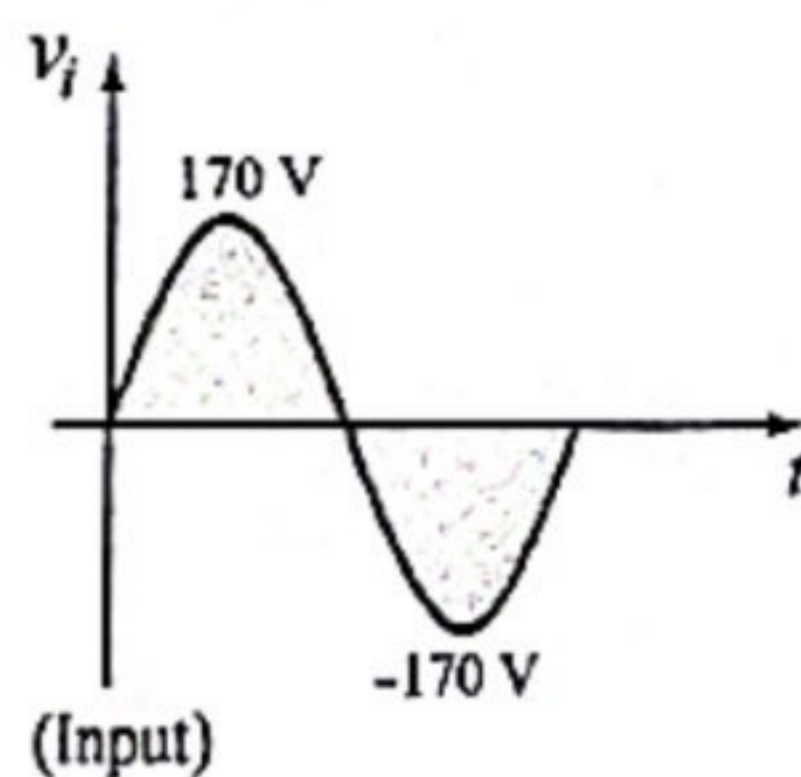
Total Marks: 40

Answer any four (4) questions. All parts of each question must be answered sequentially.
 Figures in the right margin indicate full marks.

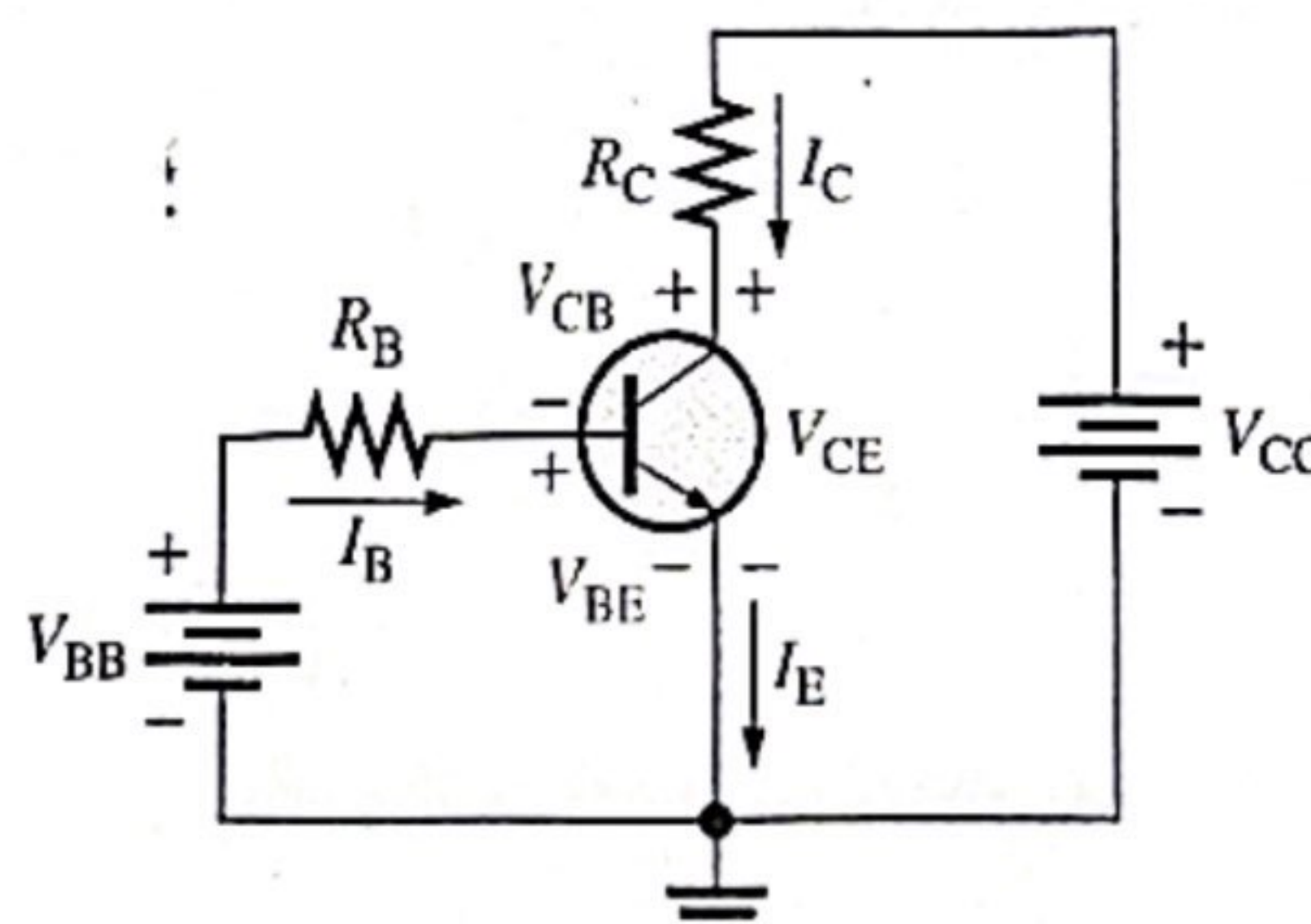
- ✓ 1. (a) Explain the necessity and role of the doping process during the formation of an extrinsic semiconductor. 2
- (b) Describe the formation process and structure of a *p*-type semiconductor material with appropriate diagrams and illustrations. 4
- (c) Calculate the currents I_1 , I_2 , and I_{D2} for the circuit in the figure below. 4



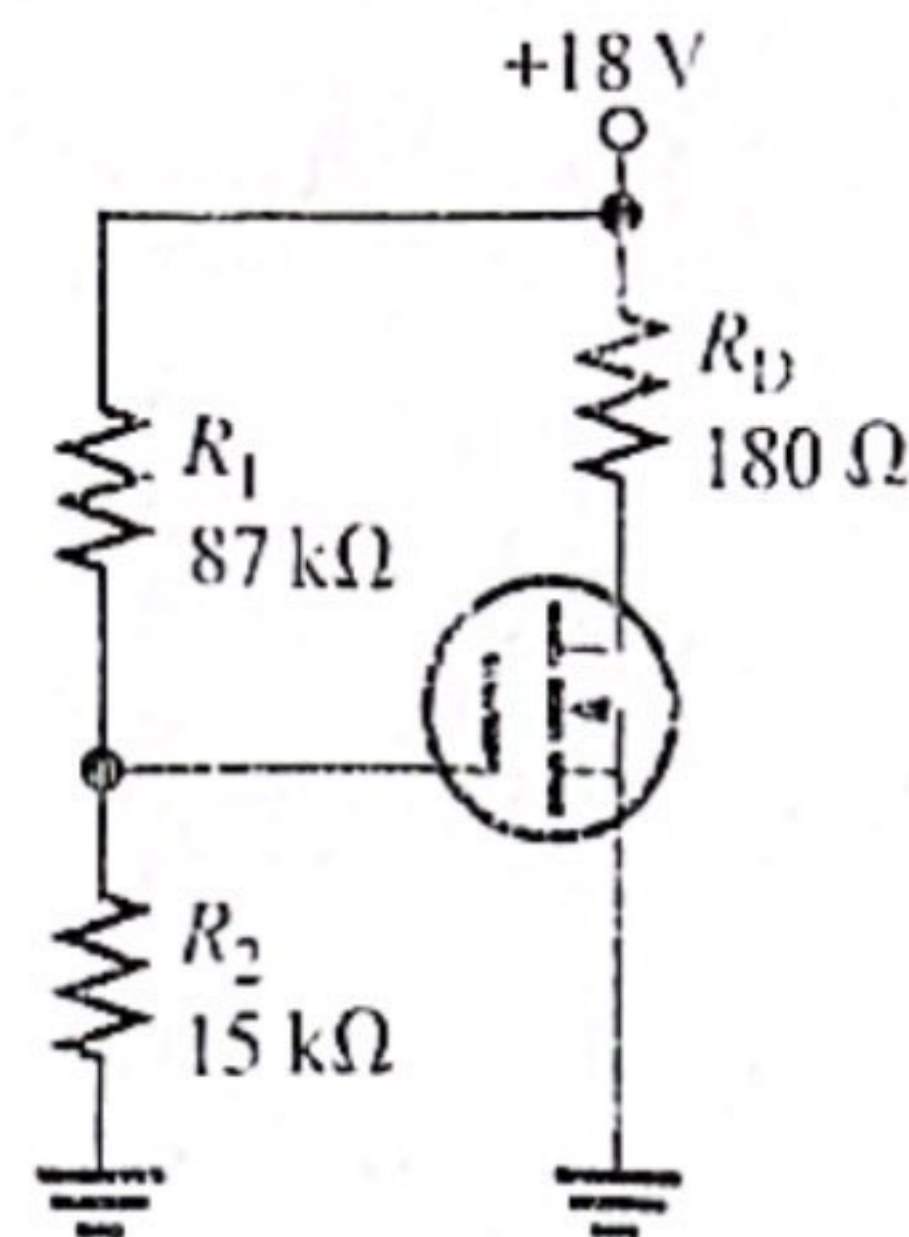
- ✓ 2. (a) Sketch the characteristics curve of a *p-n* junction diode and identify key elements. 2
- (b) Describe the working principle of a bridge rectifier circuit while converting alternating current to direct current. Use appropriate diagrams and waveforms. 4
- (c) Determine the output voltage v_o and sketch the output waveform for the network below. Also determine the dc voltage level (V_{dc}) available for the output. 4



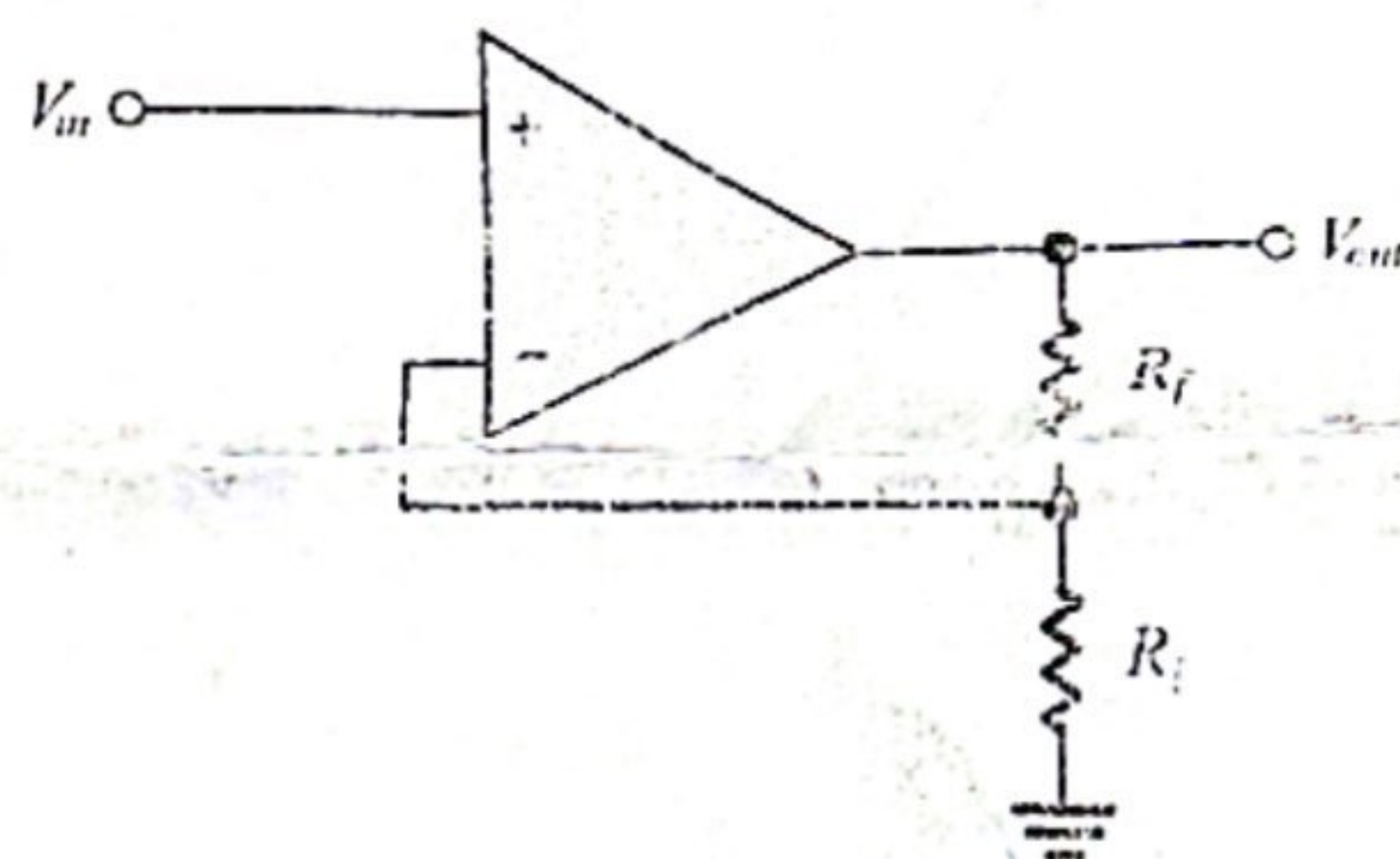
- ✓ 3. (a) Develop a relationship between the current amplification factors α and β of a BJT. 2
- (b) Discuss an overall comparison between BJT and MOSFET including their advantages, disadvantages and applications. 4
- (c) Calculate I_B , I_C , V_{CE} , V_{CB} in the transistor circuit shown in the figure below. The transistor is made of silicon and has $\beta = 180$. Given that $V_{BB} = 9V$, $V_{CC} = 15V$, $R_B = 12K\Omega$, $R_C = 120\Omega$. 4



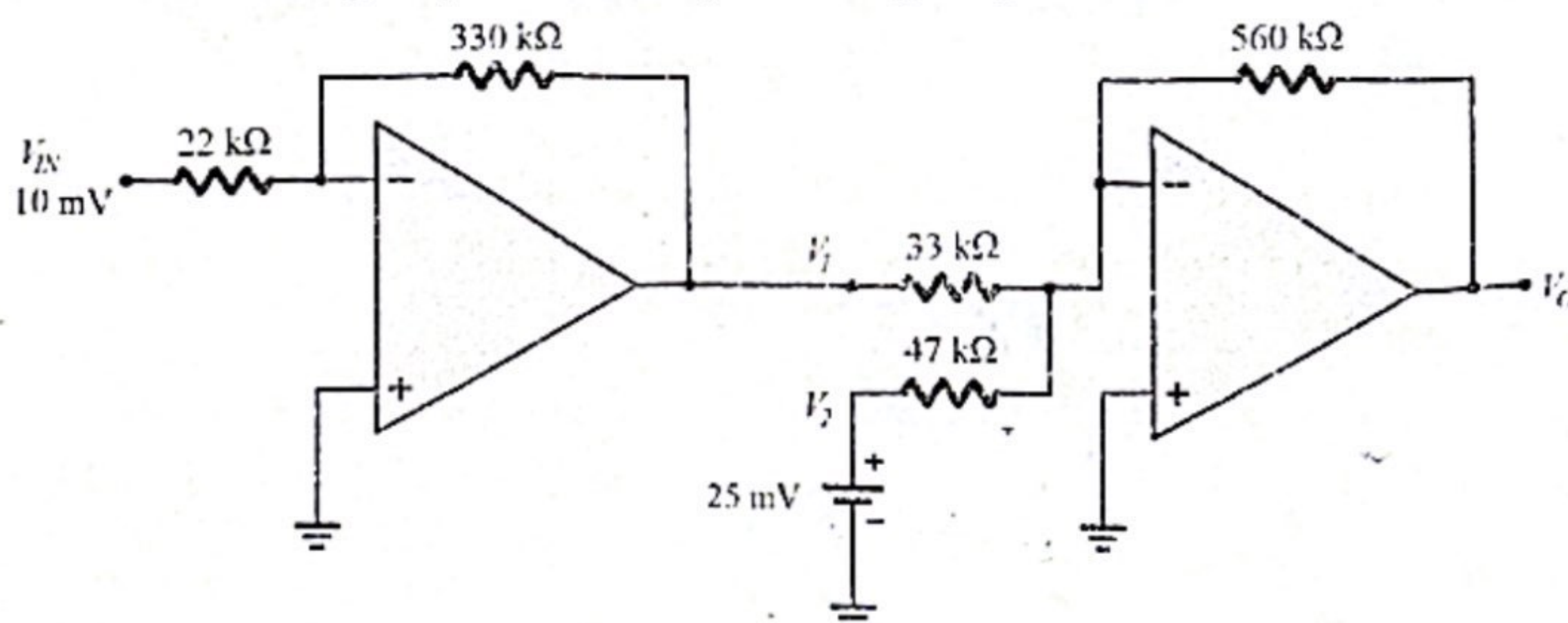
4. (a) Explain what *threshold voltage* and *pinch-off voltage* signify in the context of a MOSFET. 2
 (b) Briefly explain the basic structure, operation and drain characteristics of an *n*-channel enhancement type MOSFET with proper diagrams and illustrations. 4
 (c) Determine V_{GS} and V_{DS} for the E-MOSFET circuit in the figure below. Assume this particular MOSFET has minimum values of $I_{D(on)} = 250 \text{ mA}$ at $V_{GS} = 6 \text{ V}$ and $V_{GS(th)} = 2.5 \text{ V}$. 4



5. (a) Sketch the symbol and the internal block diagram of an operational amplifier. 2
 (b) Explain the ideal characteristics (at least 6) of an operational amplifier. 4
 (c) Develop the output voltage (V_{out}) and voltage gain (A_v) equations of the circuit below. Use appropriate notations and diagrams. 4



6. (a) Recall the function of the voltage follower and summing amplifier. 2
 (b) Develop the output voltage equation of an integrator circuit made using an operational amplifier. Use appropriate notations and diagrams. 4
 (c) Calculate the node voltage V_i and the output voltage V_o for the network below. 4



Metropolitan University
Department of Computer Science and Engineering
Spring Semester Final Examination – 2024
Program: B.Sc. in CSE; Batch: 58 (All Section)
Course: STA 215: Basic Statistics and Probability

Time: 2 Hours

Marks: 40

Answer any **four** questions. All parts of each question must be answered sequentially.

Figures in the right margin indicate full marks.

✓ The chairman of a marketing department at a large private university undertakes a study to relate starting salary (y) in thousands after graduation for marketing majors to grade point average (GPA) in major courses. To do this, records of 10 recent marketing graduates are randomly selected. The GPA (x) and the corresponding starting salary were as follows:

GPA (x)	3.26	2.60	3.35	2.86	3.82	2.21	3.47	3.28	2.54	3.25
Salary (y) in thousands	33.8	29.8	33.5	30.4	36.4	27.6	35.3	35.0	26.5	33.8

- a) Find the equation of the estimated regression line. 6
- b) Give an interpretation of the value of slope in your estimated regression line. 1
- c) Find the point prediction of starting salary corresponding to the GPA 3.95. 1
- d) Comment on the reliability of your answer in part (c). Give a reason, in context, for your answer. 2

- ✓ 2. a) Find the number of arrangements of the letters of the word 'REGRESSION' using all the letters in each arrangement. 2
- b) State the important properties of Binomial distribution. Give a practical example where binomial distribution would be appropriate to describe the situation. 2
- c) In a community, the probability that a newly born child will be a boy is $\frac{2}{5}$. Among the 5 newly born children in that community, what is the probability that: 6
- i) all five boys. 0.01024
 - ii) at least two boys. $0.33696 / 0.66304$
 - iii) at most two boys. 0.68256

✓ 3. In a training scheme for young people, the average time taken for each group to reach a certain level of proficiency was measured. The data are shown below in the table:

Age x (years)	16	17	18	19	20	21	22	23	24	25
Average time y (hours)	12	11	10	9	11	8	9	7	6	8

- a) Find S_{xx} , S_{yy} and S_{xy} . 6
- b) Use your answers in part (a) to calculate the Pearson's Correlation Coefficient (r). 1
- c) Describe and interpret the relationship between average time and age. 2
- d) Give a reason whether or not the answer in part (b) supports a linear relationship between x and y . 1

82.5
32.9
11.5

✓ 4.

Cotinine is a chemical that is made by the body of nicotine which is found in cigarette smoke. A doctor tested the blood of 12 patients, who claimed to smoke a packet of cigarettes a day, for continue. The results, in appropriate units, are shown below. [You may use $\sum x^2 = 724961$]

Patient	A	B	C	D	E	F	G	H	I	J	K	L
Cotinine level, x	160	390	169	175	125	420	171	250	210	258	186	243

a) Find the standard deviation of the level of cotinine in a patient's blood. 3

b) Find the upper and lower quartiles of these data. 2

A doctor suspects that some of his patients have been smoking more than a packet of cigarettes per day. He decides to use $Q_3 + 1.5(Q_3 - Q_1)$ and $Q_1 - 1.5(Q_3 - Q_1)$ to determine if any of the cotinine results are far enough away from the upper quartile to be outliers.

c) By drawing boxplot identify which patient(s) may have been smoking more than a packet of cigarettes a day. Show your working clearly. 5

✓ 5.

a) State the important properties of Normal distribution. Name a random variable, which has a normal distribution. 3

A psychologist gives a student two different tests. The first test has a mean of 80 and a standard deviation of 10 and the student scored 85.

b) Find the probability of scoring 85 or more on the first test. 3

The second test has a mean of 100 and a standard deviation of 15. The student scored 105 on the second test.

c) Find the probability of a score of 105 or more on the second test. 3

d) State, giving a reason, which of the students two test scores was better. 1

6. a) A contestant on quiz show is asked to choose one of three doors. Behind one of the doors is the star prize of a sports car, but behind each of the other two doors there is a toy car. The contestant chooses one of the three doors. The host then opens one of the remaining two doors and reveals a toy car. The host then asks the contestant if they want to stick with their first choice or switch to the other unopened door. State what you would recommend the contestant to do in order to have the greatest probability of winning the sports car. Show your working clearly. 5

b) Tariq either catches the bus to university or he walks to university. On any day that he goes to university, the probability that he catches the bus is 0.4. On any day he goes to university, the probability that Tariq is late for university when he catches the bus is 0.25. And the probability that Tariq is late for university when he walks is 0.05.

i) Complete the probability tree diagram. 3

ii) Find the probability that, on any day he goes to university, he is late for university. 2

Z value	0.33	0.50	0.75	1.00	1.25	1.50	1.75	2.00	2.26
P value	0.6293	0.6915	0.7734	0.8413	0.8944	0.9332	0.9599	0.9772	0.9881

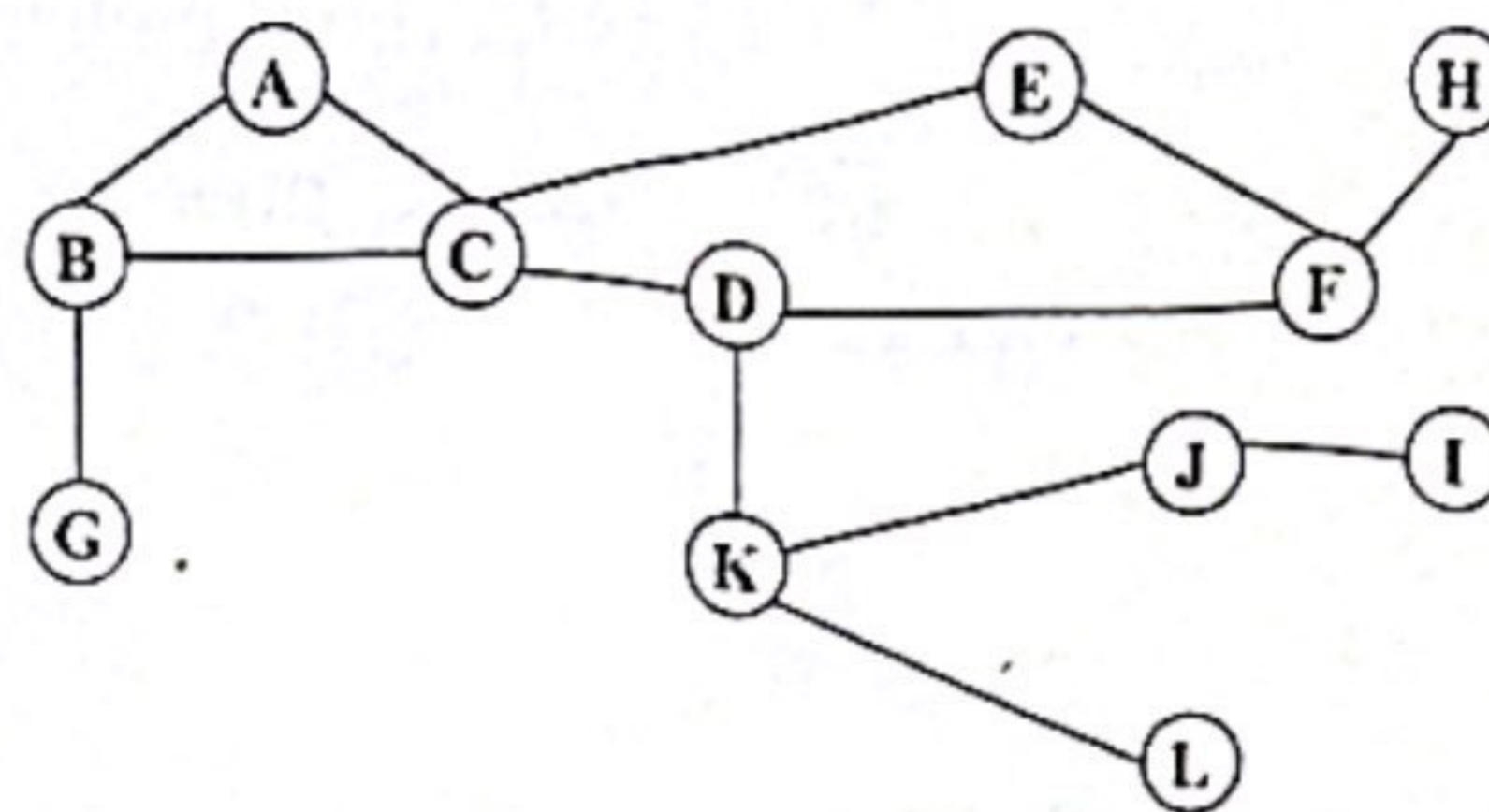
Metropolitan University, Sylhet
Department of Computer Science and Engineering
Spring Semester Final Examination - 2024
Program: B.Sc. in CSE (2nd Year 1st Semester); Batch: 58
Course: CSE 133: Data Structure

Time: 2 Hours

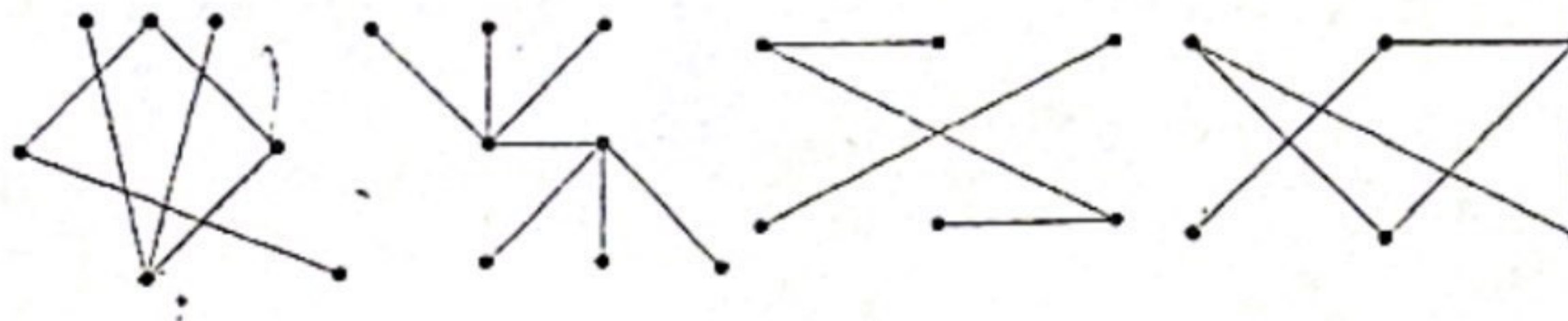
Marks: 40

Answer any **four** sets of questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

- ✓ 1. Traverse the following graph in BFS and DFS order. Show the implementation process using stack or queue. Assume that A is the starting node. (10)



- ✓ 2. (a) Consider the following graphs and determine whether each is a tree or not, justifying your answer. Based on the examples, prove the statement "Every tree is a graph but every graph is not a tree". (4)



- (b) Given the following array representation of a binary tree, draw the resultant tree. (3)

20	15	35	12	17	21	39		16	18		36	45
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- (c) Traverse the resultant tree in Inorder, Preorder, and Postorder. (3)

- ✓ 3. (a) Construct a Binary Tree from the given In-order and Post-order traversal orders.
In-order: D, B, F, E, A, G, C, L, J, H, K
Post-order: D, F, E, B, G, L, J, K, H, C, A (5)
- (b) Answer the following questions based on the resultant tree from 3(a). (5)
- What is the height of this tree? Which vertices are leaves?
 - Which vertices are children and siblings of H?
 - Which vertices are ancestors of F and descendants of C?
 - Find the level of each vertex in the rooted tree.
 - Identify the sub-tree rooted at C.

4. (a) Imagine you are Harry Potter and have heard of a lost magical spellbook hidden deep within the Forbidden Forest. The forest is a maze of paths, dense foliage, and secret clearings. Some paths are blocked by magical creatures, and others lead to dangerous areas. Starting your journey at the forest's edge, your mission is to find the magical spellbook. Based on this scenario, answer the following questions: (5)

1. Which traversal method, BFS or DFS, would you use to determine if there is a path from the entrance to the spellbook? Justify your answer.
2. Which traversal method, BFS or DFS, would you use to find the shortest path to the spellbook and ensure you encounter the least amount of danger? Justify your answer.
3. What data structures would you use in both the scenario?

- (b) Convert the given Postfix expression into a Infix expression using Stack, and show each step: $ABCD \wedge E - FGH * + \wedge * + I -$ (3)

- (c) Evaluate the following expression using a Stack: $623 + -382 / + * 2 \wedge 3 +$ (2)

5. (a) Consider this Circular Queue.

MANGO	ORANGE			APPLE
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FRONT is 4, REAR is 1 (assume 0-based indexing). In this circular queue, run the following operations and show the queue after each operation. (5)

(1) ~~ENQUEUE PAPAYA~~, (2) ~~ENQUEUE ORANGE~~, (3) ~~DEQUEUE~~, (4) ~~ENQUEUE BANANA~~, (5) ~~DEQUEUE~~, (6) ~~ENQUEUE PEAR~~, (7) ~~DEQUEUE~~

- (b) Consider a Singly Linked List with a cycle. Using Floyd's Cycle-Finding Algorithm, identify and remove the cycle to make the list acyclic. Explain the algorithm and provide a step-by-step process using an example linked list. (5)

- 6/ (a) Suppose the following list of letters is inserted in order into an empty Binary Search Tree: 39, 27, 45, 18, 29, 40, 54, 9, 21, 28, 36, 59, 10, 19, 65, 60 (6)

1. Construct the Binary Search Tree and show all steps.
2. Delete 59 from the tree and find the final tree. Show all steps.
3. Is the resultant binary tree a full binary tree, a complete binary tree, or both? Justify your answer.

- (b) Answer the following questions: (4)

1. Suppose you have a list of phone numbers that you frequently look up and insert numbers into. Which data structure would be best suited for the list? Justify your answer.
2. Suppose someone is sending integer numbers to you once a minute for an indefinite amount of time. Which data structure would you use to store the numbers? Justify your answer.

Metropolitan University Sylhet
 Department of Computer Science & Engineering
 Spring Semester Final Examination – 2024
 Program: BSc in Computer Science & Engineering
 Batch: 58th, Academic Year: 2022 (Spring)
 Course Title: **Principles of Economics & Entrepreneurship Development**
 Course Code: **GED-213**

Time – 2 Hours

Full Marks: 40

[Answer any Four (4) sets of questions. Figures in the right margin indicate full marks.]

1. (a) Compare the characteristics of firms operating in perfect competition, monopolistic competition, and monopoly market structures. Illustrate your answer with real-world examples. 6
- (b) “Netflix has reached approximately 260.28 million subscribers in 2023, despite facing fierce competition from streaming rivals like Amazon, HBO, Disney Plus, and Hulu, as well as competition from cable.” (The Economic Times, 2023) Explain why more firms have entered the streaming-video market and how their economic profit will likely change. 4
2. (a) Distinguish between real GDP and Nominal GDP. 3
- (b) Discuss the possible effects of demand-pull and cost-push inflation on both consumers and businesses. 4
- (c) Outline suitable policy measures to minimize the impact of inflation to maintain macroeconomic stability. 3
3. (a) Define entrepreneurship. How can one differentiate an entrepreneurial firm from any other types of firms? What are the three primary reasons people become entrepreneurs? 4
- (b) What evidence do you have that weaken the myth that entrepreneurs are born not made? 2
- (c) In general, what effects does entrepreneurship have on economies and societies around the world? 4
4. (a) What is the opportunity gap? List the four essential characteristics of an opportunity and how to identify an opportunity. Give an example of the social trends that are currently affecting how individuals behave and set their priorities. 4
- (b) Explain how “solving a problem” can create a business opportunity. Provide a relevant real-world example of a problem-solved business opportunity. 3
- (c) Design a step-by-step plan for organizing a focus group to generate new business ideas. 3
5. (a) Why is feasibility analysis important for a new entrepreneur? List the contents of a concept statement. 4
- (b) Identify the characteristics of an attractive industry with reference to Barringer and Michel Porter. 3
- (c) Briefly describe the two primary issues to consider when conducting an organizational feasibility analysis. 3
6. (a) Define business plan and specify two primary reasons for writing a business plan. 4
- (b) Differentiate among a summary business plan, an operational business plan, and a full business plan. 4
- (c) Give two examples of Standard Business Model. 2